Forecasting: Theory and Practice

Winter Semester 2026/27

Prof. Dr. Sebastian Lerch, sebastian.lerch@uni-marburg.de
Department of Mathematics and Computer Science, Philipps-Universität Marburg



Overview

0. Organizational Matters

1. Introduction, Examples and Applications

2. Prediction Spaces, Calibration and Sharpness

References

0. Organizational Matters

Organization

Overview:

- *Specialization module* – 9 CP module
- Primary target audience: M.Sc. students of the department's programs (in particular Data Science, Mathematics, Business Mathematics)

What should you already know?

- Necessary: Introductory lecture to statistics (e.g., *Elementare Stochastik*)
- Recommended: *Machine Learning*, and some programming experience (ideally in `Python` or `R`, e.g., Internship Stochastics)
- Optional, but a plus: advanced statistics lecture(s) (e.g., *Measure and Integration Theory / Probability Theory, Statistics and Statistical Learning, Computational Statistics, ...*)

Lectures

The lecture takes place in person on

- Thursday, 16:15 - 17:45, HS VI (03A11)
- Friday, 10:15 - 11:45, HS VI (03A11)

Lecture will be recorded (but not live-streamed) and uploaded videos will be available on ILIAS, potentially for a limited period of time only.

Office hours: No fixed dates, please make an appointment by email, in which you briefly describe your case.

Exercises

There will be 6–7 bi-weekly exercise sheets, on which you can work as a team of up to 3 students.

The exercises will be a mix of theoretical and programming exercises.

We will implement a peer-feedback mechanism via ILIAS:

- Solutions are submitted as anonymized PDFs via ILIAS.
- In the following week, your team will be randomly assigned solutions by 2 other teams for providing feedback and comments.
- The feedback will then be visible in anonymized form to the team that submitted the solution.
- We will randomly check a subset of the submitted solutions and the submitted feedback.

Exercises and criteria for exam admission

Criteria for exam admission:

- Submission of solutions to all exercise sheets.
- Submission of feedback to all exercise sheets by other teams, which are assigned to your team.
- You are allowed one missed submission of either a solution to an exercise sheet or a feedback submission.
- If your submitted solution or feedback is subject to a random check and fails an assessment (e.g., in case of severely incorrect/incomplete solutions, or incorrect/inappropriate feedback and comments):
  - You will get one opportunity to revise your submitted solution / comments.
  - Your subsequent submissions will be checked regularly and carefully. Another failed check will result in failing to qualify for exam admission.

Exercises: Organizational details

Solutions and feedback have to be submitted via ILIAS until Wednesdays at 12:00 (only by one person designated by the team).

A tutorial (by Julius Günther) will take place on

- Wednesday, 14:15 - 15:45, SR II A3 (03A21)

starting on October 28, 2026.

Exemplary solutions to the exercises (for which feedback is to be submitted in the respective week) will be presented in the tutorial.

Exams

Exams: Oral or written exam, depending on the number of participants.

- Decision within the first few weeks of the semester
- exam registration via MARVIN
- Regulations, registration periods etc.: `https://www.uni-marburg.de/de/fb12/fachbereich/dekanat/pruefungsbuero/anmeldung-zu-modulpruefungen`

Prospective exam dates:

- First exam: 25.02.2027, HS B
- Second exam: 18.03.2027, HS B

Course material and updates

- all relevant material, information, news, and dates will be announced (only) via ILIAS, `https://ilias.uni-marburg.de/`
- Magazin > MARVIN: Kurse aller Semester > Wintersemester 2026/27 > FB12 Mathematik und Informatik > ...

Contact and communication

Contact details:

- Sebastian Lerch (lecture): sebastian.lerch@uni-marburg.de
- Julius Günther (exercises): julius.guenther@uni-marburg.de

There is a forum / discussion board which you can use

- for asking organizational questions of potential general interest
- to look for team members for the exercises,

Surveys

For interactive surveys and quizzes during the lectures, we will use PINGO, an open-source class room system, developed by the University of Paderborn.

To participate

- scan the QR code, or
- visit `https://pingo.coactum.de/` and enter access number 257256
- refresh the website to see new questions



Participation is anonymous and PINGO adheres to German/EU regulations for data processing and privacy.

Survey: Study programs

Which is your study program?

- BSc / MSc Data Science
- BSc / MSc Computer Science
- BSc / MSc Mathematics
- BSc / MSc Business Informatics
- Lehramt
- Other

Survey: Prerequisites

Which other lectures did you attend already? (multiple choice)

- Elementary Stochastics or other introduction to statistics
- Statistics and Statistical Learning
- Computational Statistics
- Other advanced lecture on probability theory or statistics
- Internship Stochastics
- Machine Learning
- Advanced lecture on Machine Learning

Survey: Programming experience

How much experience do you have with `R`?

- No programming experience at all
- No experience, but have used other programming languages
- Basic knowledge, but need to look up most basic functions
- More advanced knowledge

Literature

- These slides are partly based on (unpublished) lecture notes by Tilmann Gneiting (Karlsruhe Institute of Technology).
- References to original literature are provided throughout and serve as primary sources.
- Overview articles:
  - Gneiting and Katzfuss (2014)
  - Petropoulos et al. (2022)
  - Klein (2024)